

Neha Navarkar

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Education

New York University

M.S. in Robotics and Mechatronics Engineering — GPA: 3.7/4.0 Expected May 2026 **D Y Patil College of Engineering,**

Akurdi

B.E. in Electronics Engineering — GPA: 9.18/10

New York, NY

Pune, India

2019 – 2023

Skills

Autonomy & Robotics	ROS2, VINS-Mono SLAM, ZED 2i, A*/D*-Lite, voxel mapping, trajectory planning, sensor fusion
Robot Learning	Isaac Lab, PPO/RL, sim-to-real transfer, PyTorch, JAX, CUDA
Perception	OpenCV, YOLO, point-cloud processing, Rerun, Viser
Embedded & Hardware	FOC motor control, ESP32, Jetson, Qi wireless, UART/I2C/SPI, Fusion 360
Dev Tools	Python, C++, C, Docker, Linux, Git, Gazebo

Experience

Research Assistant — CILVR Lab (Generalizable Robotics & AI)

New York University

New York, NY

May 2025 – Present

- Built full autonomous nav stack for YOR (ZED 2i, Jetson/NUC): VINS-Mono SLAM, voxel mapping at 5 Hz, weighted A* planning, and pure-pursuit control — **12 mm odometry accuracy over 10 loops**, <1 s obstacle replanning, 16 mm end-effector deviation during whole-body motion.
- Contributing author, *YOR: Your Own Mobile Manipulator* (arXiv:2602.11150) — open-source \$<\$10K mobile manipulator platform for whole-body coordination and autonomous navigation research.

Graduate Assistant — VIP STEP & MakerSpace

New York University

New York, NY

Jan 2025 – Present

- Designed gait patterns and embedded perception modules for bipedal locomotion on uneven terrain; authored SOPs and led training for 50+ students on UR5 robotic arm operation.

Graduate Apprentice — Chassis Systems

Bosch

Chakan, India

Sep 2023 – Aug 2024

- Won 1st place at Bosch's MDP Global Championship (internal innovation competition across international teams) for a sensor-fusion analytics pipeline that reduced measurement discrepancies by 20%.
- Led end-to-end delivery of 5 high-impact digitization projects, owning requirements, implementation, and stakeholder sign-off.

Projects

Go2 RL Locomotion

Isaac Lab, PPO/RSL-RL, Python, Unitree Go2

- Trained Unitree Go2 in Isaac Lab (NYU Greene HPC) with torque-based PD control, gait-phase clock inputs, multi-term reward shaping, and per-episode friction randomization ($\mu_v \sim \mathcal{U}(0, 0.3)$) for sim-to-real robustness; bonus branch adds uneven terrain curriculum.

CPR Training Machine

Raspberry Pi 5, AK60-6 BLDC, FOC Motor Control, Python/C

- Portable AHA-compliant CPR device: slider-crank with FOC torque feedback achieves compression depth within 5% of target. Stack: C/Python, ZeroMQ IPC, Kivy GUI, OpenCV patient placement, breathing detection via current sensing.

Awards & Hackathons

Best Agentic Use — Innate Robotics Le Robot Hackathon

OpenAI-powered robotic poker player with gesture recognition for bet execution and real-time win-rate strategy adaptation.

2025